Do Moving-Average Signals Predict Next-Day Returns?

Out-of-Sample Evidence from 98 NSE Stocks

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Abstract. I test whether four technical features built from past prices, a 5-day and a 20-day moving average, the one-day return and the ratio of the two averages, predict the next-day return of 98 stocks listed on India's National Stock Exchange. A separate least squares model is estimated for each stock on daily data from June 2019 to January 2025 and evaluated on the following seventeen months, which play no part in estimation. Out of sample, the mean R^2 is -0.0155 and mean directional accuracy is 49.6%. No stock's accuracy is significantly above 50%, and Pesaran-Timmermann tests, which allow for the base rate of up days, find no stock with significant directional ability once the false discovery rate is controlled. The model's apparent edge over an always-up benchmark is smaller than the edge an uninformative forecast would have earned in the same window. The results are consistent with weak-form efficiency at the daily horizon.

Keywords: technical analysis; moving averages; return predictability; out-of-sample evaluation; directional accuracy; market efficiency; India
JEL classification: C53, C58, G14, G17

1. Introduction

Technical analysis rests on the premise that the recent path of a price says something about its next move, and moving-average rules are its most common form. In the best-known version, a short moving average that rises above a long one, a *golden cross*, is read as a signal to buy, and the reverse crossing, a *death cross*, as a signal to sell. Figure 1 shows both in Britannia Industries' share price in 2019: the 5-day average rose above the 20-day average on 28 August 2019 and fell back below it on 8 November 2019. Almost every charting platform draws these averages, and many retail investors trade on them.

The academic evidence on them is mixed and has weakened over time. Lo et al. (2000) put the visual patterns of technical analysis on a statistical footing and find that several carry incremental information about returns. Brock et al. (1992) found that moving-average and trading-range rules applied to the Dow Jones Industrial Average from 1897 to 1986 produced returns that standard null models of prices could not explain. Sullivan et al. (1999) re-examined the same universe of rules with a bootstrap that corrects for data snooping and found that the best rule's performance did not carry over into the following decade. In their survey of 95 modern studies, Park and Irwin (2007) report that most find profits until the early 1990s, but that many are affected by data snooping, ex post selection of rules and difficulties in measuring risk and transaction costs. For South Asian markets, Gunasekarage and Power (2001) report evidence of predictive ability for moving-average rules in index data for four exchanges, including the Bombay Stock Exchange. More generally, Welch

*Email: arjunprashanth6129@gmail.com. Code, data and the script that generates every number, table and figure in this paper are at https://github.com/arjunprashanth6129/price-prediction-linear-regression. An interactive summary of the results is at https://arjun-quant-projects.vercel.app/linear-regression.

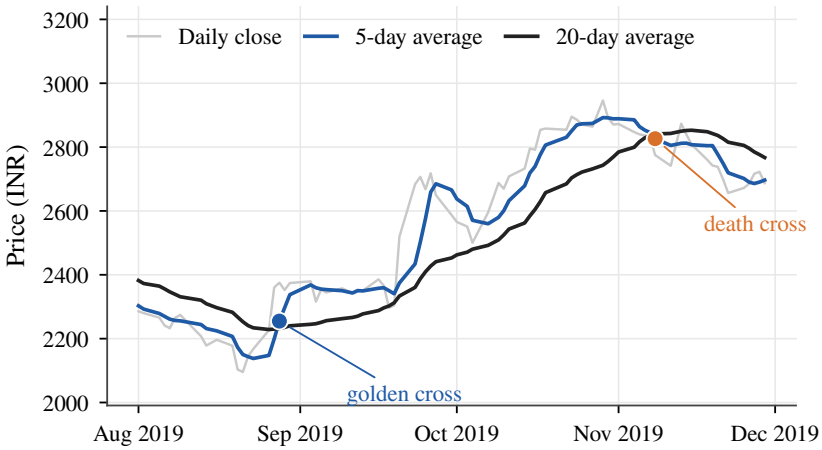


Figure 1. A golden cross and a death cross in Britannia Industries, August to November 2019. The golden cross is the day the 5-day moving average of the close rises above the 20-day average; the death cross is the day it falls back below. The ratio of the two averages, one of the four features in this study, is above one between the two crossings and below one outside them.

and Goyal (2008) show that most variables proposed as predictors of equity returns fail to beat the historical mean once they are evaluated on data not used to select them, and Campbell and Thompson (2008) formalise that comparison with the out-of-sample R^2.

This paper asks a narrow version of the question for a recent sample of Indian stocks: given only a stock's recent price history, summarised by four standard technical features, does a linear model forecast the sign and size of the next day's return more accurately than benchmarks that use no model? The null hypothesis is that the features carry no information about that return. The design choices that determine whether the answer can be trusted were fixed before any model was estimated. The split between estimation and evaluation is chronological, the features are standardised with statistics from the estimation sample only, the target is a return rather than a price, and the benchmarks against which the model is judged were chosen in advance.

The answer is negative. Across 98 stocks, the mean out-of-sample R^2 is -0.0155, and 81 stocks have a negative value. Mean directional accuracy is 49.6%. The model exceeds an always-up benchmark for 68 stocks, by +1.85 percentage points on average, but this reflects the benchmark rather than the model: up days were scarce in the evaluation window, and a forecast with no information would have exceeded the same benchmark by +2.22 points. No stock's accuracy is significantly above 50%, and the coefficient on the moving-average ratio, the variable that encodes the crossing rule, is never the largest of the four.

The paper makes three contributions. It provides out-of-sample evidence on moving-average features for a recent, broad sample of Indian stocks across three capitalisation bands. It shows how a weak always-up benchmark in a flat market can manufacture an apparent edge, and that the appropriate comparison is the expected edge of an uninformative forecast, which can be computed stock by stock. And it documents, with a worked example, why a high R^2 on price levels says nothing about forecasting ability.

I came to this question through an equity research programme at BugleRock Capital, where the contrast between fundamental analysis and technical trading came up repeatedly. Testing the most popular technical signal on data it had not seen seemed the most direct way to find out how much the evidence supports it.

Section 2 describes the data, Section 3 the method and Section 4 the results. Section 5 presents the worked example, Section 6 discusses the findings and Section 7 their limitations. Section 8 concludes.

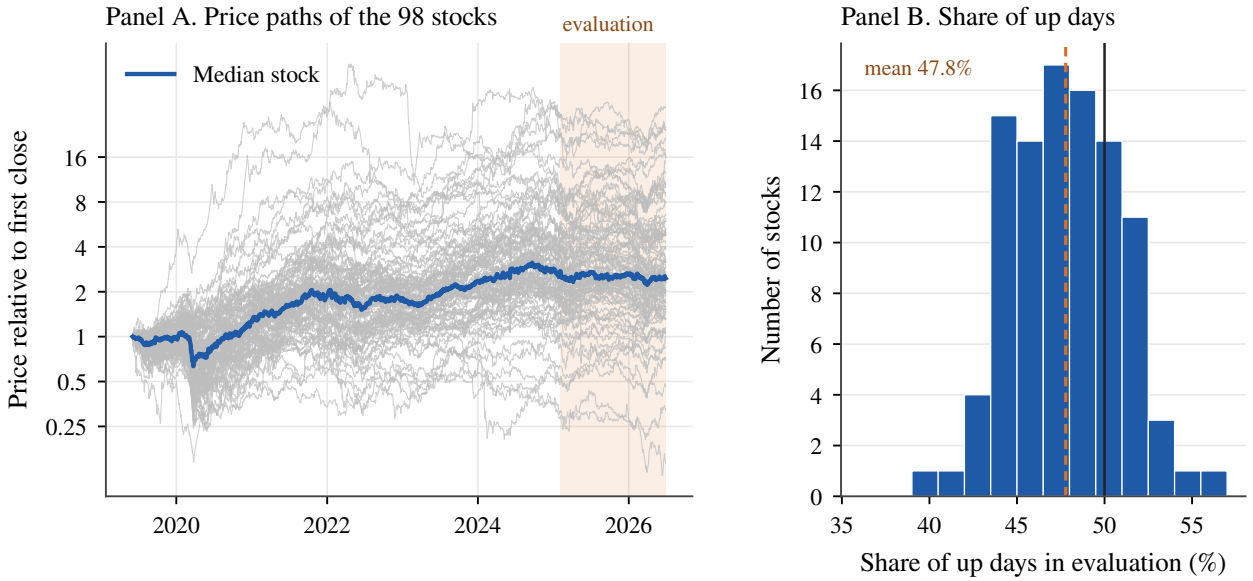


Figure 2. The sample. Panel A shows each stock's adjusted close relative to its first close, on a logarithmic scale, with the median of the stocks that have a full history in blue; the shaded area is the evaluation period. Panel B shows, for each stock, the share of evaluation days with a positive return. The black line marks 50% and the dashed line the mean.

2. Data

The universe consists of 100 companies listed on the National Stock Exchange of India, chosen to span large, mid and small capitalisation. Each stock is assigned to a capitalisation band as of June 2021. Daily closing prices from 1 June 2019 to 30 June 2026 were downloaded from Yahoo Finance and adjusted for splits and bonus issues, which would otherwise appear as large one-day losses. The sample period is fixed rather than trailing, so the dataset can be reproduced exactly; the price file is committed with the code.

Figure 2 shows the sample. Over the estimation period the median stock with a full history rose by a factor of 2.73, but over the evaluation period the median stock ended almost where it began, with a change of −0.06%. That flat evaluation window matters later, because it leaves up days in the minority for most stocks (Panel B) and so weakens the simplest benchmark.

Two stocks returned no usable history over the period and are excluded, leaving 98 stocks. Of these, 77 traded on every session in the period. The other 21, mostly companies listed after June 2019, use the history available to them. After the features are constructed, the median stock has 1,731 daily observations, of which 1,384 are used for estimation. The evaluation sample has a mean length of 345.4 trading days, from 225 for the most recent listing to 347.

Price adjustment does not correct for demergers. Two demergers fall inside the evaluation window: Raymond on 14 May 2025 and Tata Motors on 14 October 2025. Each produces a one-day price fall that is a corporate action rather than a market move. Both stocks are retained and the issue is noted in Section 7.

3. Method

3.1 Features and target

Let P_t denote a stock's adjusted close on day t. The four features use closes up to and including day t:

$$ma5_t = \frac{1}{5}\sum_{i=0}^{4} P_{t-i}, \qquad ma20_t = \frac{1}{20}\sum_{i=0}^{19} P_{t-i}, \qquad r_t = \frac{P_t}{P_{t-1}} - 1, \qquad ratio_t = \frac{ma5_t}{ma20_t}. \tag{1}$$

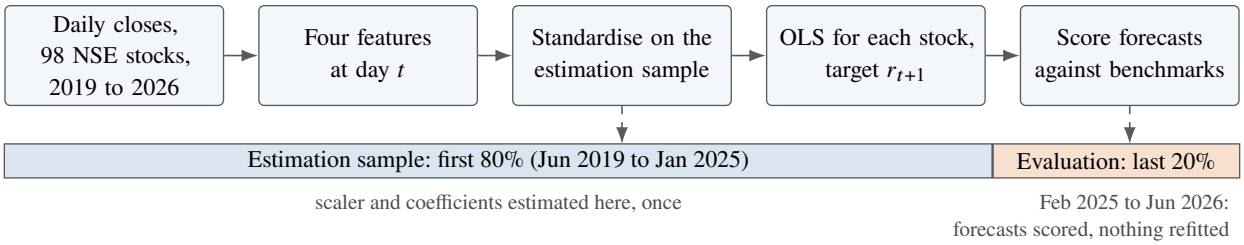


Figure 3. Research design. Features are computed from closes up to day t and the target is the return from t to $t+1$. The split is chronological and is applied separately to each stock; the dates shown are those of a stock with a full history. All parameters, including the standardisation, are estimated on the estimation sample and held fixed over the evaluation sample.

The ratio is above one when the short average is above the long one, the golden-cross state, and below one in the death-cross state. Its distance from one measures how far apart the averages are. Expressing the crossing condition as a continuous variable allows the regression to assign it a coefficient, which a binary crossing indicator that fires a few times a year would not support. The target is the next day's return, $r_{t+1} = P_{t+1}/P_t - 1$. Section 5 explains why the target is a return rather than a price.

3.2 Estimation

For each stock i, the model

$$r_{i,t+1} = \beta_{i0} + \sum_{k=1}^{4} \beta_{ik}\, \tilde{x}_{ik,t} + \varepsilon_{i,t+1} \tag{2}$$

is estimated by ordinary least squares, where $\tilde{x}_{ik,t}$ is feature k standardised to zero mean and unit variance. Standardisation makes the coefficients comparable: each β_{ik} is the change in the forecast return for a one standard deviation change in the feature. The model has five parameters per stock, which limits the scope for overfitting on roughly 1,400 estimation observations, and its coefficients can be read directly. Estimation uses scikit-learn (Pedregosa et al., 2011).

3.3 Sample split and the flow of information

The first 80% of each stock's observations form the estimation sample and the remaining 20% the evaluation sample, in date order (Figure 3). Two sources of look-ahead are excluded. A random split would place later observations in the estimation sample and judge the model on earlier ones; it would also evaluate the model on rows that are near duplicates of estimation rows, because *ma20* on consecutive days shares 19 of its 20 inputs. And the standardisation parameters are computed from the estimation sample alone and then applied to both samples, so the mean and variance of the evaluation period do not enter the transform. Nothing is re-estimated during the evaluation sample.

3.4 Evaluation

Three statistics are computed on each stock's evaluation sample of n_i days. The *out-of-sample* R^2 compares the model with a constant forecast,

$$R^2_{OS,i} = 1 - \frac{\sum_t \left(r_{i,t+1} - \hat{r}_{i,t+1}\right)^2}{\sum_t \left(r_{i,t+1} - \bar{r}_i\right)^2}, \tag{3}$$

where $\bar{r}_i$ is the mean return over the evaluation sample. A negative value means that the model's squared error exceeds that of the constant. This reference is stricter than the one used by Campbell and Thompson (2008),

the estimation-sample mean, because $\bar{r}_i$ is not known when the forecast is made. *Directional accuracy* is the share of days on which forecast and realised returns have the same sign,

$$\mathrm{DA}_i = \frac{1}{n_i} \sum_t \mathbf{1}\left[\operatorname{sign} \hat{r}_{i,t+1} = \operatorname{sign} r_{i,t+1}\right], \tag{4}$$

with a day counted as up when its return is strictly positive. The *edge* is accuracy minus the accuracy of a rule that always predicts a rise, which equals the share b_i of up days in the evaluation sample: $\mathrm{Edge}_i = \mathrm{DA}_i - b_i$. The always-up rule needs no model and is the natural benchmark for a long-only investor, but its accuracy depends on the market: it is high in a rising window and low in a flat or falling one.

The edge must therefore be read against the edge of a forecast with no information. A forecast that predicts a rise with probability q, independently of the outcome, has expected accuracy $qb_i + (1-q)(1-b_i)$. For $q = \frac{1}{2}$ this is 50% for every stock, so

$$\mathbb{E}\left[\mathrm{Edge}_i \mid \text{uninformative forecast}\right] = \tfrac{1}{2} - b_i. \tag{5}$$

Whenever up days are in the minority, a forecast with no information beats the always-up rule by construction.

3.5 Inference

For each stock, a two-sided exact binomial test compares directional accuracy with 50%. Because 98 hypotheses are tested together, the procedure of Benjamini and Hochberg (1995) is applied to control the false discovery rate at 5%. The binomial test ignores the base rate of up days. The test of Pesaran and Timmermann (1992) allows for it by comparing accuracy with the accuracy expected if forecasts and outcomes were independent. With p_y the share of up days, p_x the share of days on which the model forecasts a rise and $P^* = p_y p_x + (1-p_y)(1-p_x)$,

$$\mathrm{PT}_i = \frac{\mathrm{DA}_i - P^*}{\sqrt{\widehat{\mathrm{Var}}(\mathrm{DA}_i) - \widehat{\mathrm{Var}}(P^*)}} \xrightarrow{d} N(0,1) \tag{6}$$

under the null of no directional ability, with the variance estimators given by Pesaran and Timmermann (1992). The statistic is undefined when a model forecasts the same sign on every day.

4. Results

Table 1 summarises the evaluation across the 98 stocks.

4.1 Explained variance

The mean out-of-sample R^2 is −0.0155 and the median −0.0090. All values lie between −0.130 and +0.042, and the 17 positive values are all small (Figure 4, Panel A). There is no tail of stocks on which the model performs well. A negative R^2 for the typical stock means that the fitted model forecasts next-day returns less accurately than a constant equal to the evaluation-period mean. This is the outcome expected when the regressors carry no information: the coefficients are fitted to noise in the estimation sample, and the noise does not recur.

4.2 Directional accuracy and the benchmark

Mean directional accuracy is 49.6%, with a cross-sectional standard deviation of 3.1 percentage points and a range of 41.3% to 55.3%. A t-test of the cross-sectional mean against 50% gives $t = -1.20$, and even this overstates the evidence, because all stocks share one evaluation window and their errors are correlated.

Table 1. Out-of-sample performance across 98 stocks

Panel A. Fit of the return forecast	
Mean R^2_{OS}	−0.0155
Median R^2_{OS}	−0.0090
Range	−0.130 to +0.042
Stocks with $R^2_{OS} > 0$	17 of 98
Panel B. Direction of the forecast	
Mean directional accuracy	49.6%
Range of directional accuracy	41.3% to 55.3%
Mean share of up days (always-up benchmark)	47.8%
Mean edge over always-up (pp)	+1.85
Expected edge of an uninformative forecast (pp)	+2.22
Stocks with a positive edge	68 of 98
Panel C. Per-stock inference, 5% level	
Accuracy significantly above 50% (binomial)	0
Accuracy significantly below 50% (binomial)	7
Significant after Benjamini-Hochberg	0
Significant Pesaran-Timmermann statistic	5
Panel D. The moving-average ratio	
Stocks where `ma_ratio` has the largest coefficient	0 of 98
Positive coefficient	61 of 98
Mean absolute coefficient	0.0016

Statistics are computed on each stock's evaluation sample and averaged across stocks where stated. R^2_{OS} is defined in equation (3). The always-up benchmark's accuracy equals the share of up days. The expected edge of an uninformative forecast is $\frac{1}{2} - b_i$ from equation (5), averaged across stocks. Panel C counts stocks significant in two-sided tests at the 5% level; the Pesaran-Timmermann statistic is defined for 97 stocks. pp denotes percentage points.

The model beats the always-up rule for 68 of 98 stocks, by +1.85 percentage points on average (Figure 4, Panel B). That figure says more about the benchmark than about the model. Over the evaluation window the median stock's price changed by −0.06% and only 49 of 98 stocks rose, so up days were in the minority for 73 stocks and the always-up rule averaged only 47.8%. By equation (5), a forecast with no information would have beaten it by +2.22 points on average, more than the model achieved. Figure 5 makes the comparison stock by stock. The model's edge tracks the edge an uninformative forecast would have earned, with a correlation of 0.55 across stocks, and the points scatter around the 45-degree line rather than above it.

4.3 Per-stock inference

With a mean of 345.4 evaluation days, the standard error of a single stock's accuracy under the null of 50% is about 2.7 percentage points. In the binomial tests, no stock is significantly above 50% and 7 are significantly below; roughly 2.5 would be expected in each tail by chance, and none of the 7 survives the Benjamini-Hochberg correction. The Pesaran-Timmermann statistic, defined for 97 stocks (one model forecast a fall on every day), has a median of 0.37. It is significant at 5% for 5 stocks, all positive, which is close to the number expected by chance among 97 tests, and for none after the same correction.

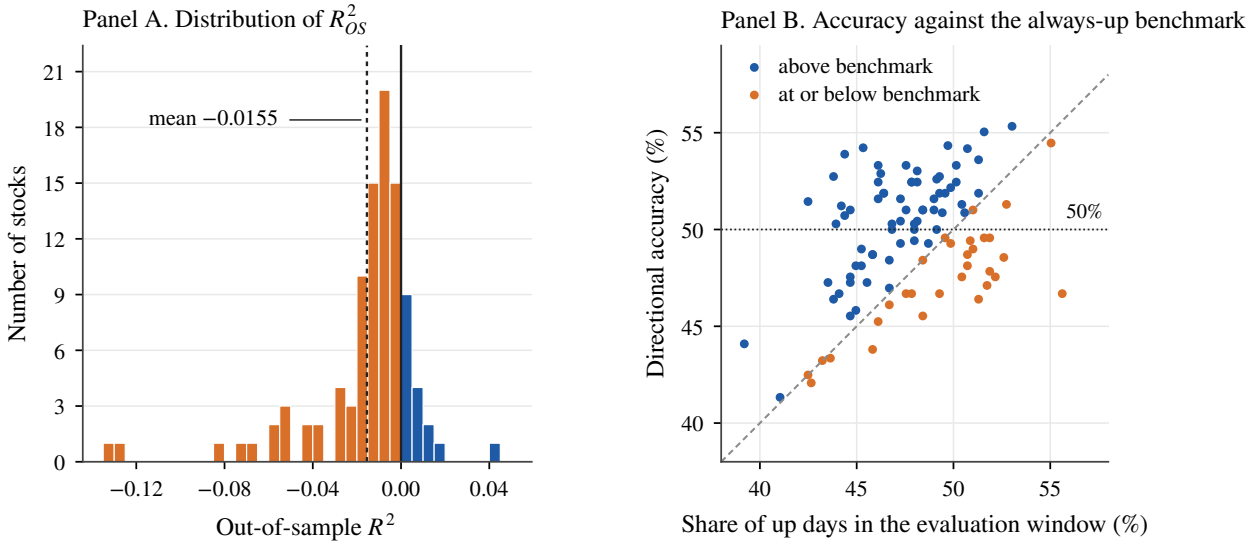


Figure 4. Explained variance and directional accuracy. Panel A shows the distribution of R^2_{OS} across stocks, with values above zero in blue; the solid line marks zero and the dashed line the mean. Panel B plots each stock's directional accuracy against the share of up days in its evaluation window, which is the accuracy of the always-up rule. Points above the dashed 45-degree line beat that rule; the dotted line marks 50%.

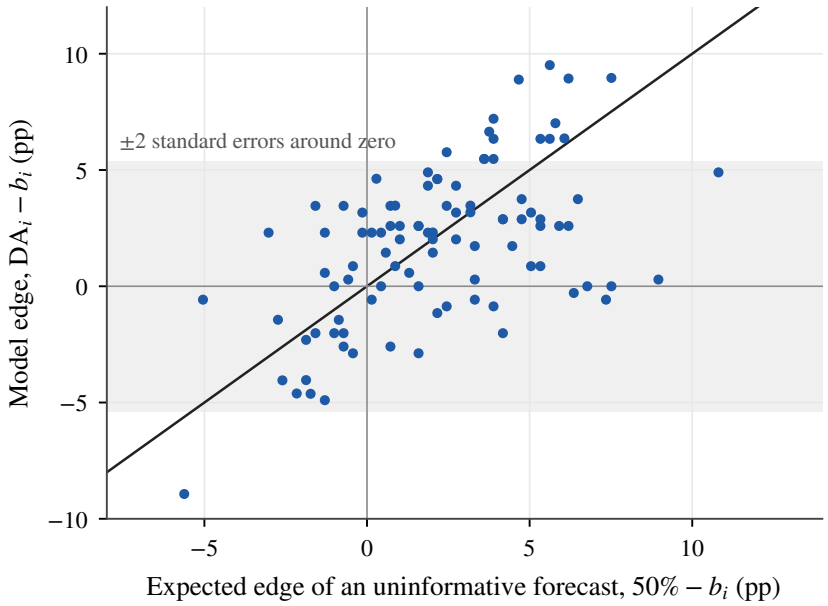


Figure 5. The model's edge against the edge of an uninformative forecast. Each point is a stock. The horizontal axis is $\frac{1}{2} - b_i$, the expected edge over the always-up rule of a forecast that predicts a rise with probability one half; the vertical axis is the model's realised edge. The solid line is the 45-degree line on which the two are equal. The shaded band is two standard errors of a single stock's accuracy around zero.

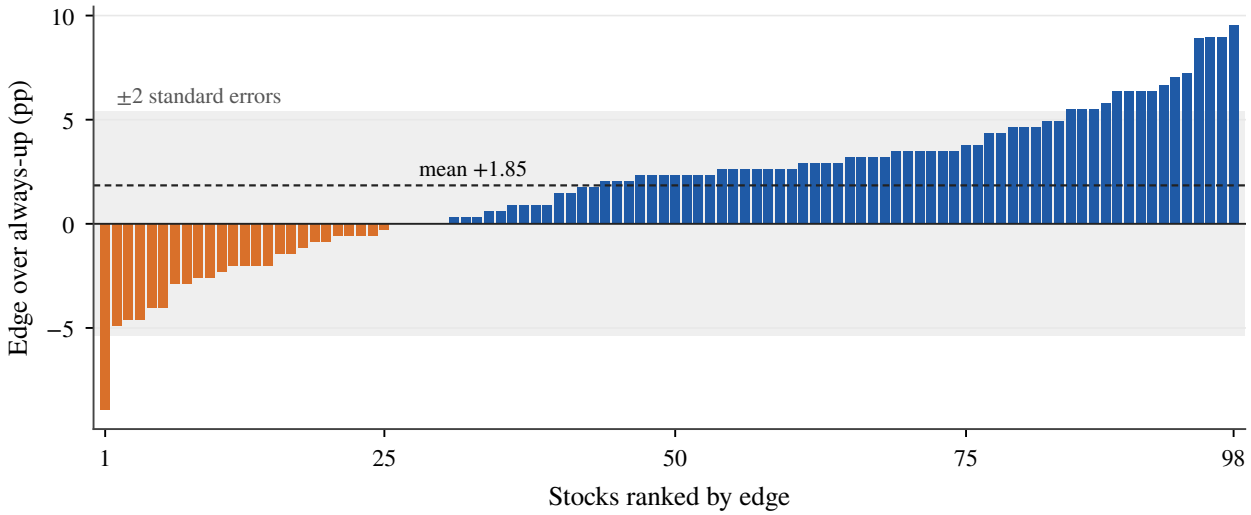


Figure 6. Each stock's edge over the always-up rule, ranked from lowest to highest, in percentage points. Blue bars are positive and orange bars negative. The shaded band is two standard errors of a single stock's accuracy around zero, and the dashed line is the mean edge.

A count of large edges seems to point the other way: 16 stocks have an edge outside a band of two standard errors around zero, 15 above and 1 below (Figure 6). That count is misleading, because the band is centred on zero while the edges are centred on +1.85 points for the reason given above. The largest edges also tend to coincide with weak benchmarks rather than with good fits: four of the five largest occur with a negative R^2_{OS} (Appendix Table 4).

4.4 The moving-average ratio

The ratio has the largest standardised coefficient for none of the 98 stocks; the largest is *ma5* for 58, *ma20* for 38 and the one-day return for 2. Its mean absolute coefficient is 0.0016, so a one standard deviation change in the ratio moves the forecast daily return by about 0.16 percentage points. The sign is positive for 61 stocks and negative for 37. A sign test against an even split gives $p = 0.02$, so the positive majority is unlikely to be chance alone, but it has no forecasting value: the coefficients are small, and the accuracy results above show no benefit. Figure 7 shows the coefficient for every stock in Panel A and compares the four features in Panel B. The mean absolute coefficients on *ma5* and *ma20*, 128.1 and 126.5 basis points per standard deviation, are about eight times the 16.2 on the ratio, but they are large mainly because they offset each other.

The individual coefficients should be read with care. The ratio is a deterministic function of *ma5* and *ma20*, which are themselves highly correlated. Under this degree of multicollinearity, least squares can assign large offsetting coefficients to correlated regressors with little change in the fitted values, so the individual coefficients have high variance. The conclusions of this paper rest on out-of-sample accuracy, which does not depend on how the fitted values are divided among the regressors.

4.5 Results by capitalisation

Hong et al. (2000) find that momentum is stronger among smaller stocks and those followed by fewer analysts, so Table 2 and Figure 8 report the results by capitalisation band. Small capitalisation stocks have the largest mean edge, but their mean accuracy is the same as that of large capitalisation stocks. The difference in edge comes entirely from a lower share of up days among small caps, not from more accurate forecasts. Mean

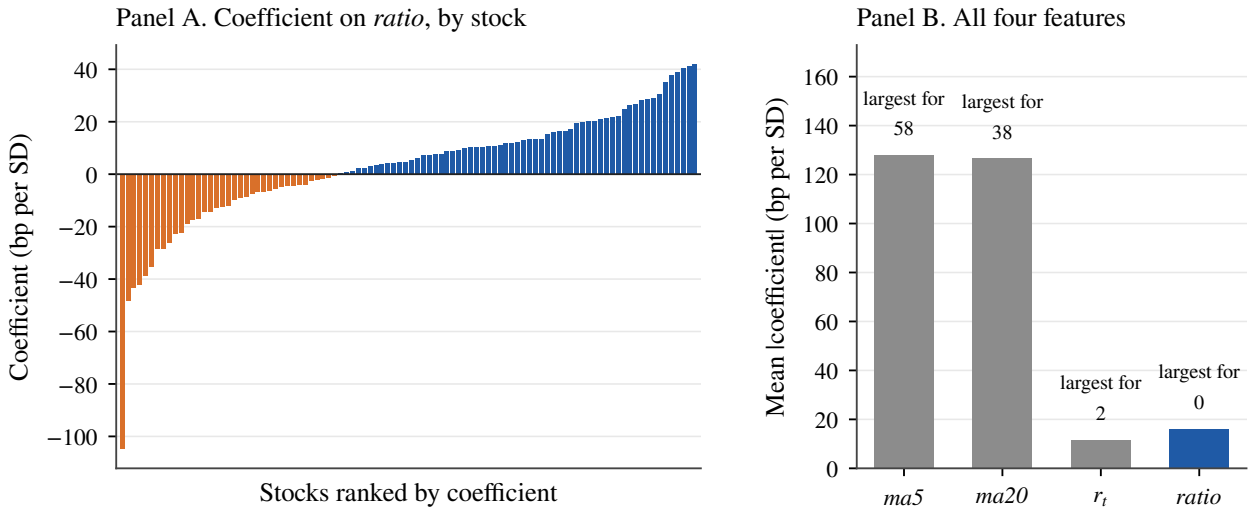


Figure 7. The standardised coefficients, in basis points of daily return per standard deviation of the feature. Panel A shows the coefficient on the moving-average ratio for each stock, ranked, with positive values in blue. Panel B shows the mean absolute coefficient of each feature across stocks and the number of stocks for which that feature has the largest coefficient.

Table 2. Results by capitalisation band

Band	n	Accuracy (%)	Up days (%)	Edge (pp)	Mean R^2_{OS}
Large	55	49.8	48.5	+1.31	−0.0130
Mid	17	48.9	47.2	+1.68	−0.0235
Small	26	49.8	46.7	+3.09	−0.0153
All	98	49.6	47.8	+1.85	−0.0155

Bands are assigned as of June 2021. Accuracy, the share of up days, the edge and R^2_{OS} are averaged across the stocks in each band. The share of up days is the accuracy of the always-up rule, and the edge is accuracy minus that share, in percentage points.

R^2_{OS} is negative in all three bands, and the dispersion within each band is larger than the differences between bands.

5. Why the target is a return: a worked example

A natural alternative design forecasts the next day's *price* rather than its return. This section shows, for Reliance Industries, why that design produces a high R^2 that is consistent with no forecasting ability. Two regressions with the same features were fitted, one to the next day's close and one to its return, both estimated on data from 1 July 2019 to 31 January 2025 and evaluated on the 347 trading days from 1 February 2025 to 26 June 2026. A third forecast, the random walk $\hat{P}_{t+1} = P_t$, requires no estimation.

The price model attains an R^2 of 0.936, the kind of figure that suggests a strong model (Table 3). The random walk attains 0.963, with a mean absolute error 27% lower, and it beats the price model on every measure in the table. The reason is in the denominator of R^2, which compares the squared forecast errors with the variance of the outcome around its mean. For a price level that variance is large relative to a single day's change. Over the evaluation sample the close had a standard deviation of INR 93.9, while the random walk's root mean squared error, the typical size of a daily move, was INR 18.1. The random walk therefore

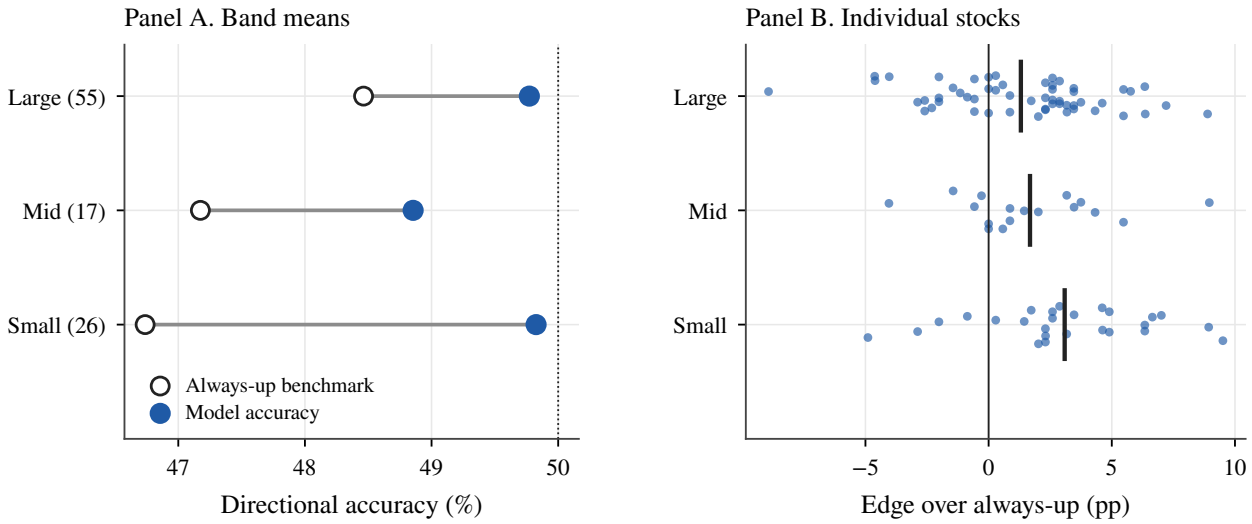


Figure 8. Results by capitalisation band. Panel A shows each band's mean always-up benchmark (hollow) and mean directional accuracy (filled); the dotted line marks 50%. Panel B shows each stock's edge, with the band mean as a vertical bar.

Table 3. Reliance Industries: level and return forecasts over the evaluation sample

	Price model	Random walk	Return model
MAE	INR 18.23	INR 13.26	0.0097
RMSE	INR 23.70	INR 18.08	0.0133
R^2 against the test mean	0.9363	0.9629	−0.0080
Directional accuracy (%)	47.0	51.6	51.0

The price model and the return model use the same four features, the same split and the same estimator; only the target differs. The random walk forecasts tomorrow's close as today's close; because it never forecasts a rise, its directional accuracy is the share of days on which the price did not rise. MAE and RMSE for the first two columns are in rupees and comparable with each other; the return model's errors are fractions of the price. R^2 is computed against the evaluation-sample mean, as in equation (3).

scores $R^2 = 1 - (18.1/93.9)^2 = 0.963$ without forecasting anything, and any forecast close to the previous close explains most of the variance of the level, whether or not it anticipates the change.

The last row of Table 3 scores direction, and there the high R^2 counts for nothing. The price model calls the next day's move correctly on 47.0% of days, worse than a coin. The random walk never forecasts a rise, so its 51.6% is simply the share of days on which the price did not rise. The return model is right on 51.0% of days against 48.4% for the always-up rule, an edge of +2.6 points, well inside the band of ±5.4 points within which an uninformative forecast falls (Section 4.3).

Figure 9 shows the two forecasts over the evaluation period. In Panel A the forecast close follows the realised close so closely that the two lines are hard to separate, because it is in effect yesterday's close. Forecasting the return removes the level, and the apparent fit goes with it. In Panel B the return forecast barely leaves zero while the realised return swings between −4.6% and +5.3%. Because the forecasts are close to zero, the residuals in Panel C are close to the realised returns themselves, and they show no structure that the model has removed.

Figure 10 plots each forecast against the outcome. The level forecasts lie along the diagonal, while the return forecasts form a narrow vertical band within 0.47% of zero. A near-constant forecast is what least

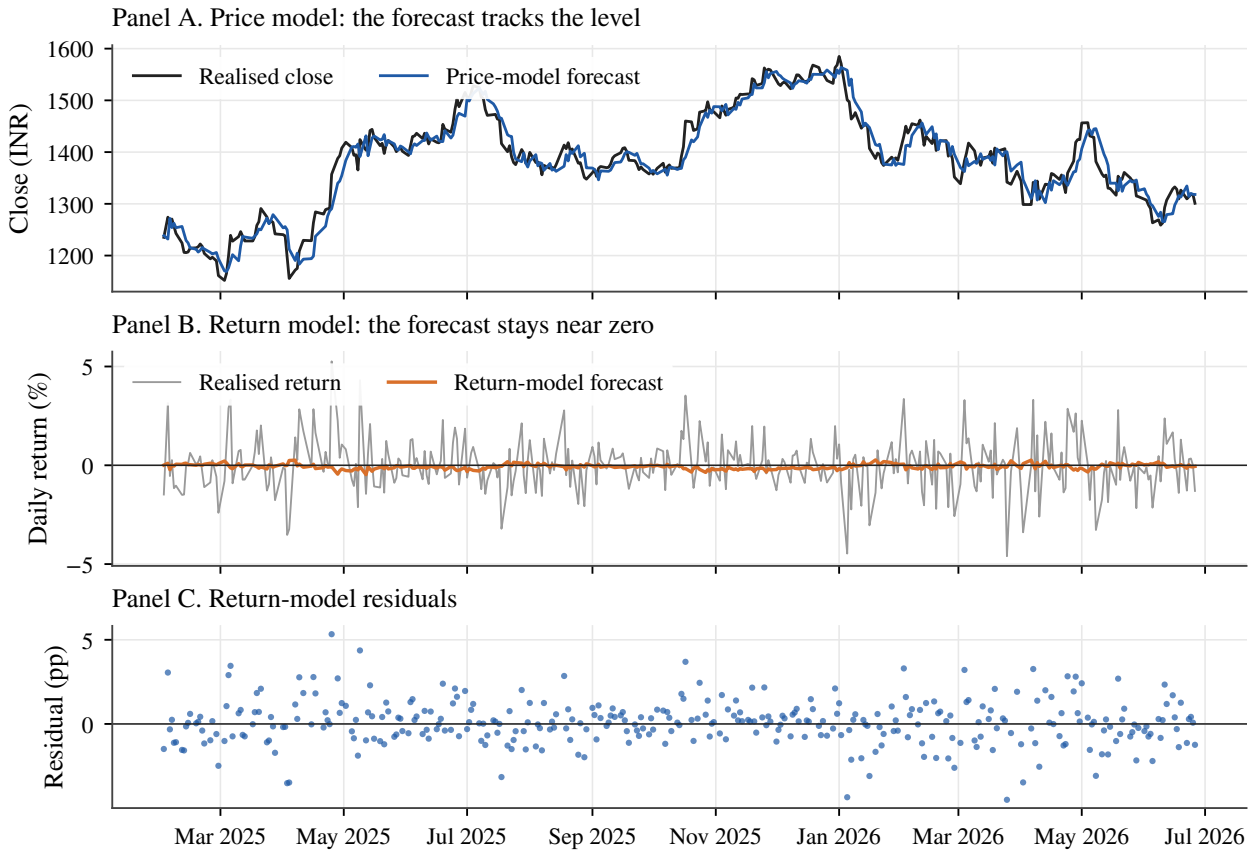


Figure 9. Reliance Industries over the evaluation period. Panel A shows the realised close and the price model's forecast, Panel B the realised daily return and the return model's forecast, and Panel C the return model's residuals, realised minus forecast.

squares produces when the regressors are uninformative: the estimated coefficients are close to zero, and the fitted value converges to the estimation-sample mean, which for a large stock's daily return is itself close to zero.

The coefficients show how each model reaches its forecast (Figure 11). The price model puts almost all its weight on *ma5*, the feature closest to today's price, so it forecasts tomorrow's price mainly by reproducing today's. In the return model the two largest coefficients, on *ma5* and *ma20*, are nearly equal and opposite. Because the two averages move together, their effects cancel on a typical day, which is how correlated regressors behave when they are fitted to noise.

Figure 12 draws the return model as a surface. With the one-day return and the ratio held at their means, the forecast is a plane in *ma5* and *ma20*, and the vertical distance from each day to the plane is the error that least squares minimises. The observations lie along a narrow diagonal band, because the two averages rarely move apart, and along that band the plane is almost flat while the realised returns scatter far above and below it. A model with real predictive power would show a plane tilted along the band, with the days close to it.

The general point is that R^2 measures variance explained relative to a chosen reference forecast. For an autocorrelated series such as a price level, the appropriate reference is the random walk, not the sample mean.

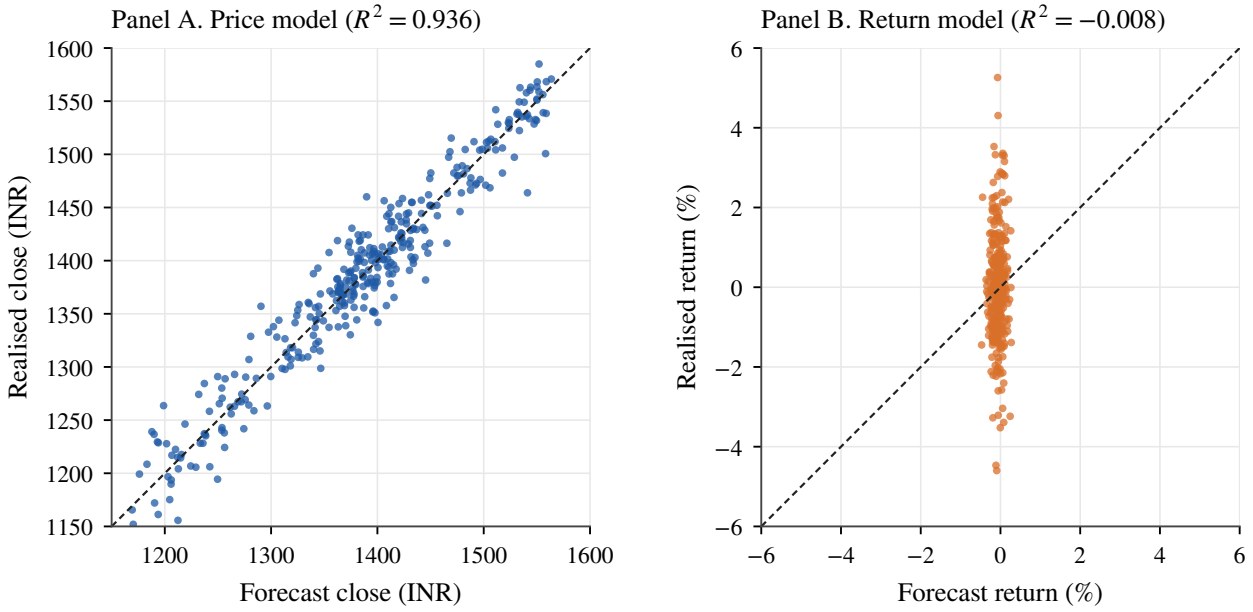


Figure 10. Forecast against realised values for Reliance Industries over the same 347 evaluation days. Panel A is the price model and Panel B the return model; the dashed line marks perfect forecasts. The level forecasts lie close to the line because prices change little from one day to the next. The return forecasts form a narrow vertical band, which indicates that they do not vary with the outcome.

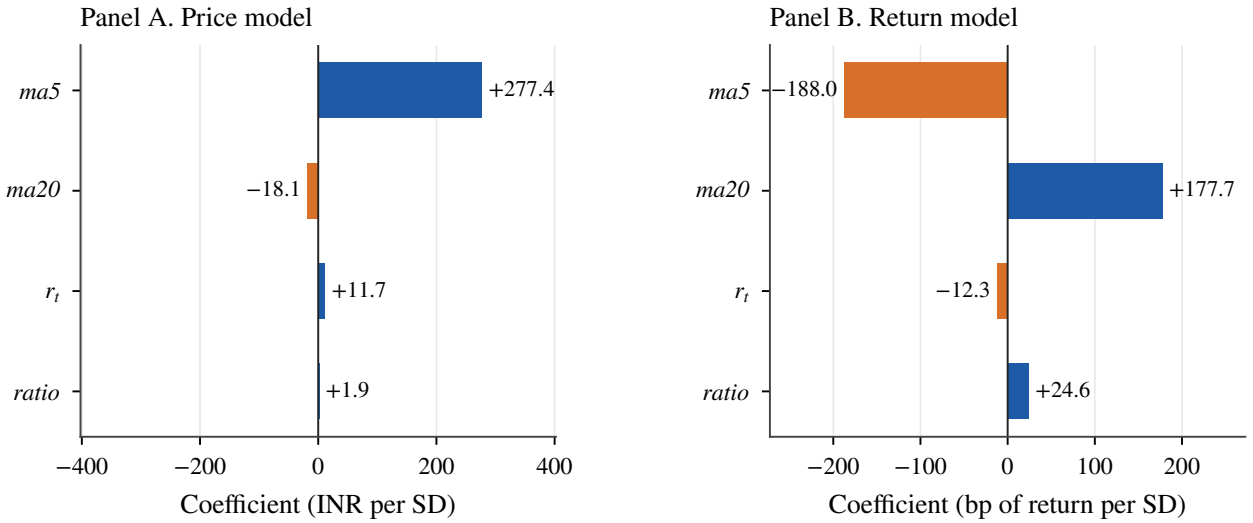


Figure 11. The standardised coefficients of the two Reliance models: rupees of forecast close per standard deviation of each feature for the price model, and basis points of forecast return per standard deviation for the return model.

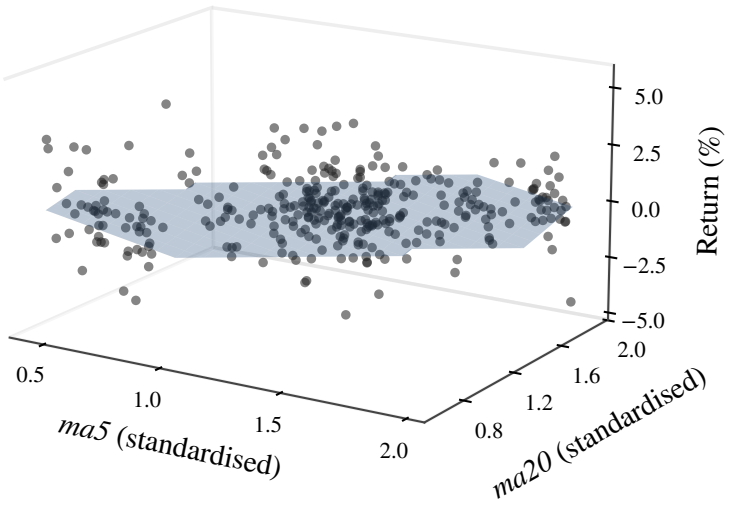


Figure 12. A two-feature slice of the fitted Reliance return model, with the one-day return and the moving-average ratio held at their means. The plane is drawn only over the range of combinations that occur in the data. The points are the realised returns of the evaluation days.

6. Discussion

The results are consistent with the weak form of the efficient market hypothesis (Fama, 1970), under which current prices reflect the information contained in past prices. A rule based on past prices that reliably preceded price increases would be traded until the return it offered was competed away. Moving averages are among the most widely computed indicators, so whatever predictability they once carried is likely to have been exploited, and the decline in the profitability of technical rules after the early 1990s documented by Park and Irwin (2007) points the same way.

The evidence is limited in a way that matters for interpretation. With about 345 evaluation days, a single stock would need directional accuracy of roughly 57.5% for a two-sided test at 5% to detect it with 80% power. Effects of one or two percentage points, which could still be economically meaningful for a large portfolio traded at low cost, cannot be detected stock by stock. Pooling across stocks helps less than the count of 98 suggests, because the stocks share a single evaluation window. What the data can rule out is a large and general effect: there is no subset of stocks, capitalisation band or feature for which the model performs reliably better than an uninformative forecast.

The results do not contradict the momentum literature. Jegadeesh and Titman (1993) document that stocks with high returns over the previous three to twelve months outperform those with low returns over the following months. That effect is measured across stocks and over monthly horizons. This paper measures the relationship between short moving averages and the next day's return for each stock separately, which is a different hypothesis.

The conclusion is also narrow in what it covers. It rules out a one-day forecast built from these four features with a linear model, on this universe and in this period, which is the horizon on which short-term technical

trading operates. It says nothing about investing on fundamentals. Profitability, reinvestment, leverage, cash flow and valuation are not inputs to the model, so the results neither support nor contradict their use.

7. Limitations

- *One evaluation window.* All 98 evaluation samples fall within the same seventeen months, February 2025 to June 2026, so the results for different stocks are correlated, and a window with different market conditions could produce a different benchmark and a different edge.
- *Specification.* The model is linear in four features. A relationship that operates through thresholds or interactions, such as a crossing that predicts returns only in calm markets, would not be detected.
- *Corporate actions.* Price adjustment does not correct the two demergers in the evaluation window (Section 2).
- *Returns and costs.* Returns exclude dividends, and no transaction costs are deducted. A small positive edge before costs would not imply a profitable strategy after brokerage, the bid-ask spread and short-term capital gains tax.
- *Universe.* Most of the stocks are large and mid capitalisation constituents, among the most widely followed Indian equities, so a null result is less surprising here than it would be among less liquid stocks. Capitalisation bands are fixed as of June 2021.

8. Conclusion

Four moving-average features in a linear model do not forecast the next day's return of 98 NSE stocks better than benchmarks that use no model. The mean out-of-sample R^2 is negative, mean directional accuracy is 49.6%, and no stock shows significant directional ability once the base rate of up days and the number of tests are taken into account. The apparent edge over an always-up rule is an artefact of a flat evaluation window, and it is smaller than the edge an uninformative forecast would have earned. Two lessons about method generalise beyond this sample: a benchmark's accuracy must be read alongside the conditions that produced it, and R^2 on an autocorrelated level is not evidence of forecasting ability.

Natural extensions are cross-sectional momentum at monthly horizons, a larger small-capitalisation sample, predictors that do not depend on price history, such as earnings revisions, and non-linear models evaluated with walk-forward validation. Replacing the ratio with an indicator for a recent crossing would remove its collinearity with the two averages and test the rule in the form in which it is used.

Use of AI tools

The research question, the design of the study, the choice of benchmarks and tests, and the interpretation of the results are my own. I used Claude Code, Anthropic's AI coding assistant, in two supporting roles. The first was as a tutor. When the work reached an idea I had not used before, such as why a shuffled split overstates accuracy when neighbouring rows share most of their inputs, I asked it to explain the idea and then checked the explanation against the data before relying on it. The second was implementation: at my direction it wrote parts of the code, including the plotting code and the script that generates this paper's numbers, tables and figures, and it helped me draft and edit the text. Working this way also taught me how to use an AI assistant well, by specifying a task precisely, asking for the reasoning behind a suggestion and not only the code, and verifying every output. I reviewed everything it produced and take full responsibility for the content.

Data and code availability

Prices, code and results are available in the project repository, https://github.com/arjunprashanth6129/price-prediction-linear-r
Running `python paper/make_assets.py` re-estimates every model, checks the results against the committed output, and regenerates all numbers, tables and figures in this paper.

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A. Largest and smallest edges

Table 4. Stocks with the largest and smallest edge over the always-up rule

Rank	Ticker	Band	Accuracy (%)	Up days (%)	Edge (pp)	R^2_{OS}	Binomial p
1	RPOWER	Small	53.9	44.4	+9.5	−0.072	0.16
2	GARFIBRES	Mid	51.4	42.5	+9.0	+0.014	0.63
3	HIMATSEIDE	Small	52.7	43.8	+8.9	−0.014	0.33
4	PAYTM	Large	54.2	45.3	+8.9	−0.084	0.23
5	CHOLAFIN	Large	53.3	46.1	+7.2	−0.005	0.24
6	GREENPANEL	Small	51.2	44.2	+7.0	+0.008	0.70
7	SUPRAJIT	Small	52.9	46.2	+6.6	+0.014	0.31
8	BALKRISIND	Large	50.3	43.9	+6.4	−0.002	0.96
			…				
91	KARURVYSYA	Small	47.6	50.4	−2.9	−0.054	0.39
92	HAVELLS	Large	45.5	48.4	−2.9	−0.011	0.11
93	DRREDDY	Large	47.8	51.9	−4.0	−0.006	0.45
94	NAVINFLUOR	Mid	48.6	52.6	−4.0	−0.059	0.63
95	FEDERALBNK	Large	47.6	52.2	−4.6	−0.025	0.39
96	ALKEM	Large	47.1	51.7	−4.6	−0.012	0.31
97	BAJAJCON	Small	46.4	51.3	−4.9	−0.068	0.20
98	LAURUSLABS	Large	46.7	55.6	−8.9	−0.130	0.24

Stocks ranked by edge, the model's directional accuracy minus the share of up days in the evaluation window, in percentage points. The binomial p-value tests accuracy against 50% and is not adjusted for multiple testing. Full results for all 98 stocks are in `results/per-stock.csv` in the repository.